CFD Pre-Processor for Impeller Passage

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Index

1. Purpose. 3

2. Scope. 3

3. Abbreviations. 3

4. General procedure: description. 3

4.1 Importing Mesh 5

4.2 Type of Analysis 5

4.3 Model Fluids 5

4.4 Boundary Conditions and Interfaces 8

4.5 Solver Control 11

4.6 Expression and Monitor 12

4.7 Summary of the setting 14

5. Acceptance criteria. 15

6. References 16

Annex 17

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1. Purpose.

The document describes a guideline to realize a model of impeller which allows you to simulate its geometry using just a single passage.

A specific example using CFX PRE of ANSYS environment is described in the Annex.

2. Scope.

This procedure is applicable to any vaned channel of centrifugal pumps, rotoric or statoric, radial, mixed or axial flow.

The procedure is applicable for the Nova Milanese and La Spezia plants

3. Abbreviations.

B.C.		*Boundary Condition*
CFD		*Computational fluid Dynamics*
D1	*[mm or in]*	*impeller eye diameter*
D2	*[mm or in]*	*maximum external diameter of the impeller*
Dh	*[mm or in]*	*shaft impeller diameter*
Eta	*[%]*	*efficiency*
H	*[m]*	*head*
LE		*leading edge of blades impeller*
Nq or Ns		*specific speed of the pump*
PS		*pressure side of blades impeller*
Q	*[m^3/h]*	*flowrate*
SS		*suction side of blades impeller*
SST		*Shear Stress Transport*
TE		*trailing edge of blades impeller*

4. General procedure: description.

The guideline provides general instructions to set up a CFD simulation of impeller using just a single passage; this model is a good simplification compared to the complete impeller model since the single passage is a portion of the entire impeller domain that is duplicated with respect to the rotation axis by a number equal to the blades present in the impeller.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

In this model the fluid processed is single-phase, therefore it allows to measure the performance of the impeller in terms of prevalence and preliminary efficiency; instead to measure the NPSH it is necessary to set another model, and it is described in the SOP:"CFD Pre-Processor for NPSH analysis".

The advantages and disadvantages compared to an impeller inserted inside a complete pump model are summarized in the Table1:

PRO	CONTRO
- the model is simple and easy to manage in the redesign phase - reduction of calculation time in the simulation - it is the good model if the flow at the impeller inlet is regular and axial and if the interaction phenomena between the impeller and the component placed downstream of it (for example volute or diffuser) are negligible.	- the effects of local unevenness of the incoming flow are difficult to be set, unless complicated boundary conditions are imposed on it. -It does not consider the effects of fluid dynamic interaction at the exit of the rotating impeller with the stator element (for example volute or diffuser) placed downstream of it. -since the presence of the disks and wear rings are not simulated, it is not possible to measure the leakage flowrate and the overall impeller efficiency, only internal power can be calculated

Table 1: advantages and disadvantages of model

The inputs needed to follow the steps described in this guideline are the mesh of the single vane impeller prepared in SOP-490.

A general discussion is presented below, with the aim to provide the user with the necessary information and recommended criteria for setting a CFD model using general purpose tools of Pre-Processor (some examples are CFX PRE, Fluent, Open Foam, Comsol Multiphysics etc.)

COPY COMPLIANT to the original document resident on the computer system at the time of printing

4.1 Importing Mesh

Usually in the mesh import phase, regardless of the preprocessing tool, it is necessary to select the type of mesh or meshing tool, if required, and the measurement scale is the orientation with respect to the orthonormal reference system. It is always necessary to verify that the rotation axis of the blade passage is always coincident with one of the axes of the orthonormal reference system.

4.2 Type of Analysis

For the model with a single blade vane, it is sufficient to set up a steady-state analysis provided that the flow that develops in the blade vane is not strongly turbulent (dynamic phenomena and separation of the flow from the wall affect negatively the convergence of the calculation).

4.3 Model Fluids

Domains are regions of space in which the equations of fluid flow or heat transfer are solved and are created from a list of Assemblies, 3D primitive regions and/or 3D composite regions that are associated with a volume of an imported mesh.

Boundary conditions can be applied to any bounding surface of a 3D region that is included in a domain.

The Solver uses the concept of domains to define the region of the fluid, the type, the properties, the morphology and the equations applied on it; are described below:

Material proprieties of fluid

The fluid processed during the simulation must be correctly defined. Depending on the calculation type that will be performed, several fluid properties must be set.
Table 2 shows the physical properties of water (and other example) necessary and sufficient to characterize the fluid in a thermo-fluid dynamic simulation:

COPY COMPLIANT to the original document resident on the computer system at the time of printing

	Example of fluid		
Material proprieties	Water	Air	Water Vapour
Thermodynamics state	liquid	gas	gas
Ref. temperature [°C]	25	25	25
Ref. pressure [bar]	1	1	0.03169
Ref. spec enthalpy [J/kg]	0	0	-13424249
Ref. spec entropy [J/(kg*K)]	0	0	10475.125
molar mass [kg/kmol]	18.02	28.96	18.02
density [kg/m^3]	997	1.185	0.022308
Specific heat capacity [J/(kg*K)]	4181.7	1004.4	1911.6
Dynamic Viscosity [kg/(m*s)]	0.0008899	0.00001831	0.0000098626
Thermal Conductivity [W/(m*K)]	0.6069	0.0261	0.01854
thermal expansivity [1/K]	0.000257	0.003356	0.003356

Table 2 : Material proprieties

Usually, each solver already has a library of chemical-physical properties of materials; if the fluid you want to model does not exist, you can create it within the library.

In our case, the fluid that will be simulated is water, as reported in the first column of the table.

Morphology of fluid

Morphology is used to describe the connectivity or distribution of the fluid, in fact it can take different forms, in the simplest case as a single continuous fluid (homogeneous and uniform in space), or as a dispersed fluid, that is a fluid that transports and enters with a dispersed solid or gaseous phase or vice versa.

Generally, the morphology used for water during single impeller vane simulation is continuous (even if inside the water there is always a gaseous and solid phase).This setting helps to reduce the computational effort.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Reference Pressure of model

In this section it is necessary to set the absolute reference pressure level of the entire physical system.

The value to be set is equal to Pref=1atm, this represents the reference pressure for all pressures imposed as boundary conditions and subsequently simulated.

Buoyancy Model

For flows where gravity is important, it is necessary to include the buoyancy term.

It is therefore necessary to insert the gravitational components in the x, y and z directions with respect to the coordinate system for which the domain is oriented.

The buoyancy model is significant for slow and multiphase flows where the natural convection term is predominant, in our case it is not necessary to insert it.

Domain Motion

In this impeller vane model, it is necessary to set the rotation of the fluid domain; therefore, it is necessary to enter the angular velocity equal to the rotation speed of the pump and define the rotation axis of the system.

Equation of Heat Transfer

The exchange of work between the impeller and the fluid occurs at almost constant temperature; in the fluid model it is therefore necessary to set the isothermal transformation condition.

Turbulence of model

Turbulence can be modelled using a variety of approaches, including DNS (Direct Numerical Simulation), LES (Large Eddy Simulation), and RANS (Reynolds-averaged Navier-Stokes) modelling.

In our case we set the turbulence model SST (Shear Stress Transport); this model is part of RANS approach, it is widely used in industry, particularly in turbomachinery.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

4.4 Boundary Conditions and Interfaces

Boundary conditions must be applied to all the bounding surface in our domain (see Figure 1); the boundary conditions can be:

- Inlet
  - Flow rate
  - Velocity distribution
- Outlet
  - Pressure
- Opening
- Wall
- Symmetry plane

Instead, in the internal connection surfaces between fluid domains, it is necessary to select the names of the two or more side of domains and the type of interface, which can be:

- General connection
- Sliding interfaces (Frozen rotor, Stage (mixing-plane), Transient rotor-stator)
- Periodic (Translational, Rotational)

Interfaces deserve a more in-depth treatment because the exchange of information through it must occur either if the two meshes are of different types (for example a hexahedral mesh with a tetrahedral one) either if the two fluid domains are rotor and stator. Thanks to numerical algorithms and multiple reference systems, the treatment of interface flows is completely implicit and conservative in terms of mass, momentum, energy, therefore without any penalty in terms of robustness or convergence speed.

In our case the only interface present in our domain is the one related to rotational periodicity; note that even if the two sides are not contiguous this type of interface allows the two surfaces to exchange information.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

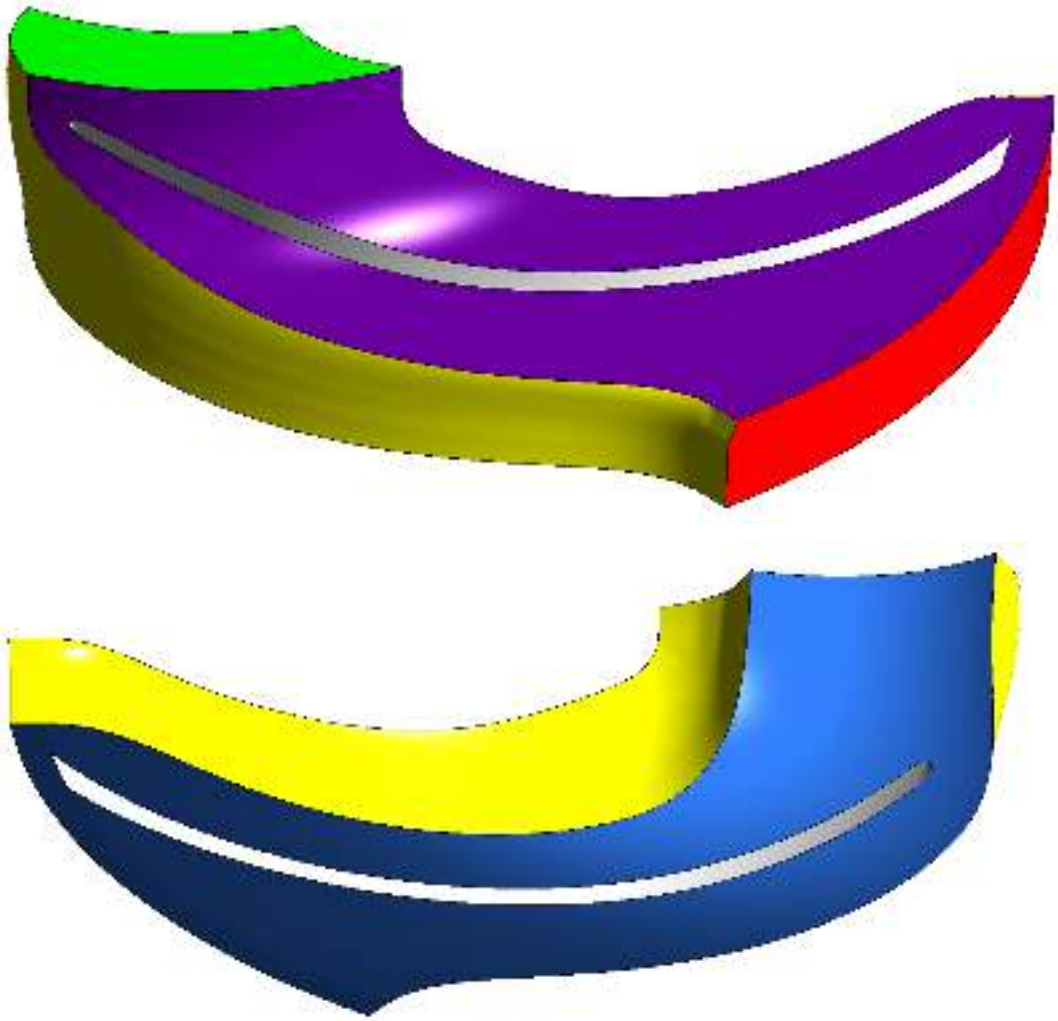


Figure 1 : Boundary condition and interface applied in domain

Table 3 shows the operating conditions and the boundary conditions to be applied on the domain patches.

Operating Condition	**Assignment B.C.**	**Color patch**
Rotational Speed [rpm]	Blade, hub, shroud	white, blue, purple
Total pressure [Pa]	Inlet	green
Mass flow rate [kg/s]	Outlet	red
Wall Roughness [micron]	Blade, hub, shroud	white, blue, purple

Table 3 : Assignment B.C.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

It is important to note that since we are preparing a simulation of a single vane of impeller, the mass flow rate will be set equal mass flow rate of the pump divided by the number of impeller blades; furthermore, to consider the wear ring leakage mass flow rate, it is possible to add it to the mass flow rate of the vane, through a preliminary calculation or from literature [1].

The roughness value depends on the type of manufacturing, that is, whether the impeller is created starting from a mechanical casting or whether it is made entirely with mechanical processing (such as by means of numerically controlled precision machines):

- From sand casting of impeller with external diameter lower than 600mm: Ra 6.3 (8 micron)
- From sand casting process of impeller with external diameter greater than 600mm Ra 12.5
- From ceramic or investment casting process or CNC machined: smooth roughness

Some detailed settings of the Inlet B.C are the subsonic flow regime, the normal flow direction and the turbulent intensity equal to 5%.

For the same reasons it is necessary to set the rotational periodicity condition between periodic surfaces shown in Table 4 with the yellow color.

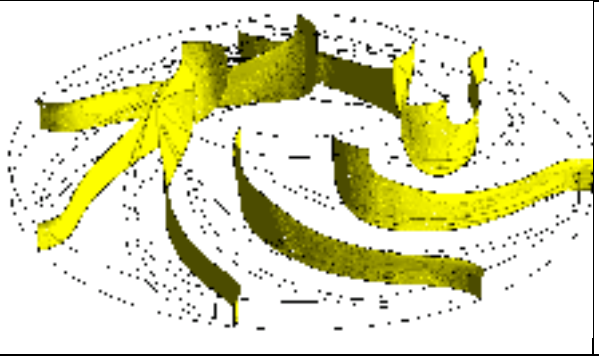
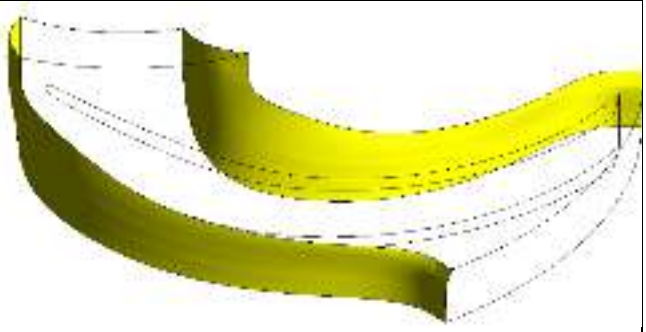
Periodicity of vanes for impeller	Periodicity of single vane
	

Table 4 : Interface rotational periodicity

COPY COMPLIANT to the original document resident on the computer system at the time of printing

4.5 Solver Control

To solve a fluid dynamics simulation, the Navier-Stokes equations are used, these are a set of coupled partial differential equations that take into account the continuity equation, the moment and the energy.

Through the use of CFD it is possible to solve these equations numerically using discretized volumes and time intervals.

The choice of the discretization of the fluid volume occurs in the previous meshing phase, and once the mesh is created the dimensions of the finite volumes are fixed, while the time interval and its management are set within the Solver, the main choice falls on the type of numerical scheme, first order, second order or second order + up-wind.

In our case, since we are using a stationary simulation, a first-order scheme is sufficient.

In the steady state cases the solver applies a false time step as a means to under-relax the equations while they iterate towards the final solution.

Convergence Control: steady state

In the setting section where convergence is checked, two approaches can be adopted:

- the first approach is based on setting a fixed number of interactions for which it is assumed that the state of convergence has arrived at a good solution.

- the second approach is instead to monitor in real time the state of convergence through the so-called residues, then a minimum residue value is set for which the simulation must converge.

In our case, it is preferable to adopt the first approach since from previous experiences it has been found that the solution arrives at an acceptable state of convergence at 1500 iterations.

The management of the time scale can be multiple and depending on the type of Solver, the technique more convenient to adopt the "Auto Timescale" function.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

4.6 Expression and Monitor

Within the solver is an "output control" panel that allows you to manage how files are written by the solver; you can control which variables will be written to the steady results files and how often the files will be created.

Monitoring data can also be displayed in real-time, so you can monitor and track the progress of the solution; this allows you to examine the results before the solution is fully converged.

Solvers usually also allow for back-up saves, which can be loaded into the post-processing section for examination.

Since all monitored variables undergo temporal fluctuations due to numerical instability (and turbulent un-stationary phenomena for unsteady simulation), it is always necessary to evaluate the convergence value by averaging the last range that falls within the ±1% convergence band.

There are different reasons for evaluating the "time-invariant" variables of a stationary simulation, since the fluctuations are due only to numerical instability.
As previously stated from previous company experiences, the iterations adopted are 1500, while the iterations for which to make the average are the last 100.

In addition to the "residues" and the default variables, it is possible to set predefined expressions that can be monitored during the simulation progress.

An example of expressions is shown in Figure 2.

Expressions

etaimp	*abs(massFlow()@inlet)*9.81*himp/pimp/10*
himp	*(areaAve(Total Pressure in Stn Frame)@outlet - areaAve(Total Pressure in Stn Frame)@inlet)/g/997*
pimp	*abs(torque_z()@blade+torque_z()@shroud+torque_z()@hub)*2980/60*2*pi/1000*

Figure 2 : Expression

In Figure 2, the monitored variables of impeller efficiency, the impeller head, the power absorbed by the impeller are shown, neglecting the friction losses on the external disks.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Many others can be inserted and monitored; the number and their complexity depend on the complexity of the pump model and the analysis activity.

For example, in addition to the main expressions, the forces with respect to the axes of the global reference system Fx, Fy and Fz can be monitored.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

4.7 Summary of the setting

The following table Table 5 summarizes the setting to be set for this SOP:

Type of Analysis	Steady State
Model Fluid	• Water at 20°C (with his properties, see Table 2) • Morphology: continuous fluid • Reference Pressure: 1 atm • Buoyancy model: not buoyant • Domain Motion: rotation equal to rotor speed • Eq. Heat Transfer: isothermal • Turbulence Model: SST
Boundary Conditions	• Inlet: Total Pressure [Pa]; subsonic flow regime; normal flow direction; Turbulence Intensity 5% @inlet domain • Outlet: Mass flow rate [kg/s] @outlet domain • Wall rotating: speed rotation [rpm]; wall roughness [micron] @hub, shroud, blade domain • Interface: rotational periodicity @periodicity side 1 and 2
Solver Control	• First-order scheme • Max. iteration convergence control: 1500 (the last 100 for the average correct value) • Timescale control: Auto Timescale (length scale option: conservative, timescale factor 1)
Expressions and monitor	• Efficiency impeller (not essential) • Head impeller • Power impeller • Fx,Fy,Fz (not essential)

Table 5 : Setting

COPY COMPLIANT to the original document resident on the computer system at the time of printing

5. Acceptance criteria.

This procedure is applicable for any geometry of impeller; indeed, it could be used for radial, semy-axial and axial impeller; furthermore, it could be used for any type of impeller, closed, semi-open and open impeller (Table 6).

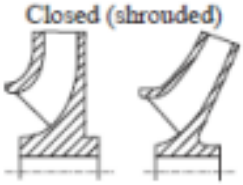
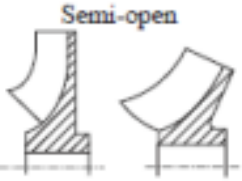
Component or feature	Radial	Semi-axial	Axial
(A) Impeller form: Characteristic: flow direction at impeller exit			
(B) Impeller type	Closed (shrouded) 	Semi-open 	Open

Table 6 : Type of geometry impeller [1]

Furthermore, more in general this guideline can be used for all the geometry vaned channels, so this guideline can be extended to diffuser of multistage pumps and bowl of vertical pumps (Table 7).

(C) Diffuser Characteristic: flow direction at diffuser inlet	Radial	Semi-axial

Table 7 : Type of geometry diffuser [1]

This simplified model is acceptable in terms of accuracy only if it is simulated with operating conditions close to the BEP of the pump, or more precisely within the range 0.8 to 1.1 with respect to the position of the BEP.

From literature [1] it is shown that in off-design operating conditions the results of Head, Efficiency and Power if compared with those of the test obtain percentage differences of even 6-8%, this does not make a redesign activity reliable on which one wants to manage the slope of the H-Q curve.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

From literature [1] it also appears that, using this simplified model during the redesign phase, a change greater than 2% of efficiency has a good chance of being captured by the CFD, while a change lower than 0.2% is insignificant since it falls within the numerical simulation error.

The setting proposed in this document refers to the background of the company on CFD analysis using a specific software [2]. These setting can be different when using different CFD software

6. References

[1] Gulich "Centrifugal Pumps" ,4th.ed.

[2] Ansys User Manual

[3] https://www.cfd-online.com/

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Annex

This section summarizes the main steps to be able to set up the impeller model using the CFX Solver tool in the ANSYS environment:

- Access the CFX-Pre setting environment (see Figure 3).

Figure 3 : CFX-Pre environment

- Import the impeller vane mesh, selecting the correct mesh tool (see Figure 4).

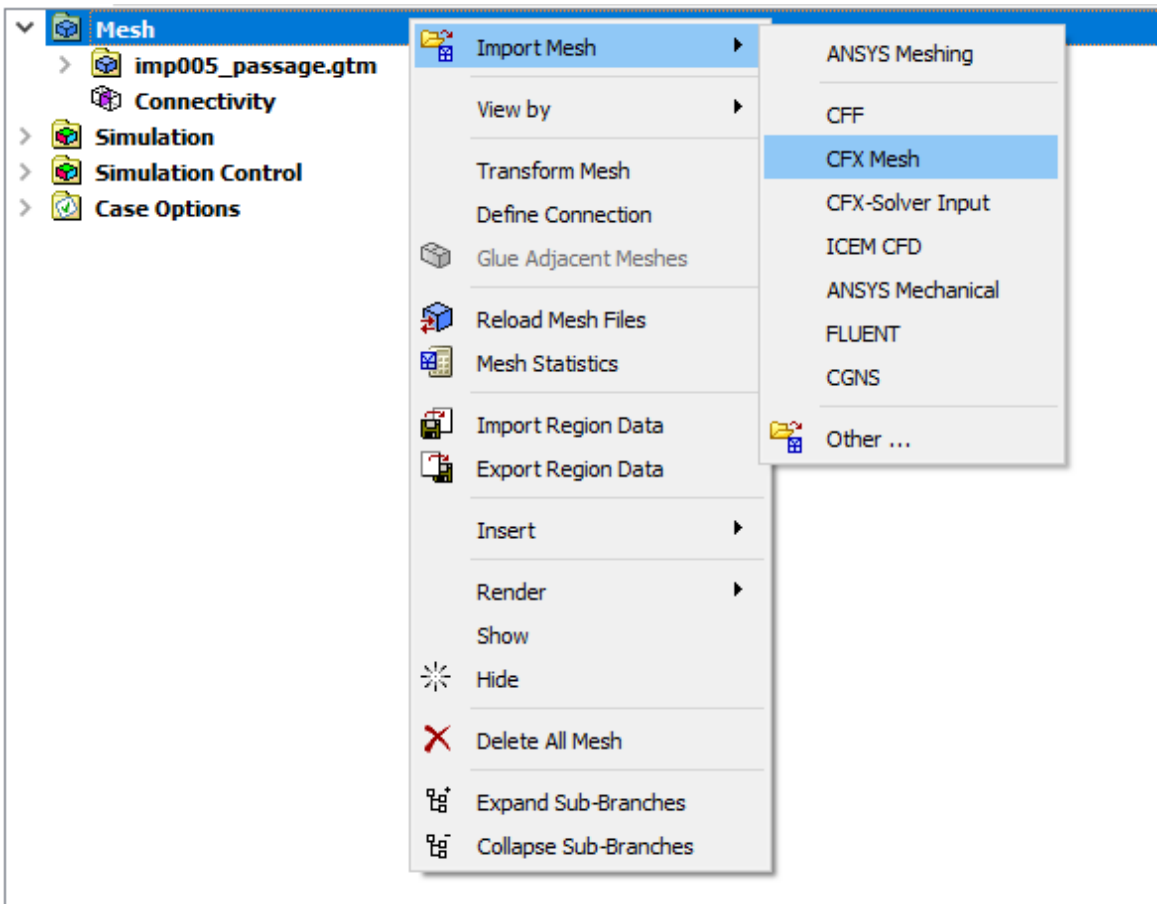


Figure 4 : Import Mesh

- Select the type of steady-state analysis (see Figure 5)

COPY COMPLIANT to the original document resident on the computer system at the time of printing

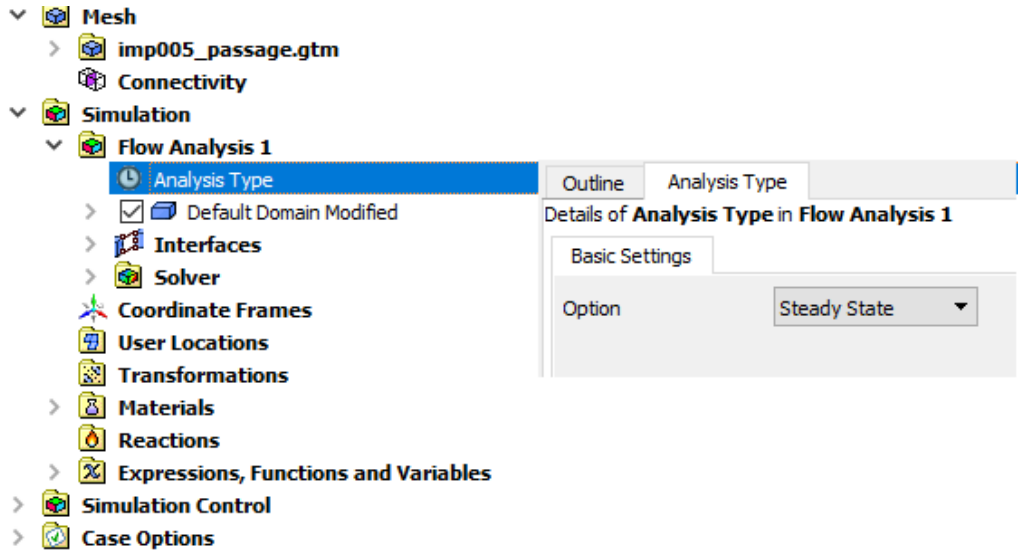


Figure 5 : Setting Steady State Analysis

- Realize the fluid model setting as shown in the figure (see Figure 6)

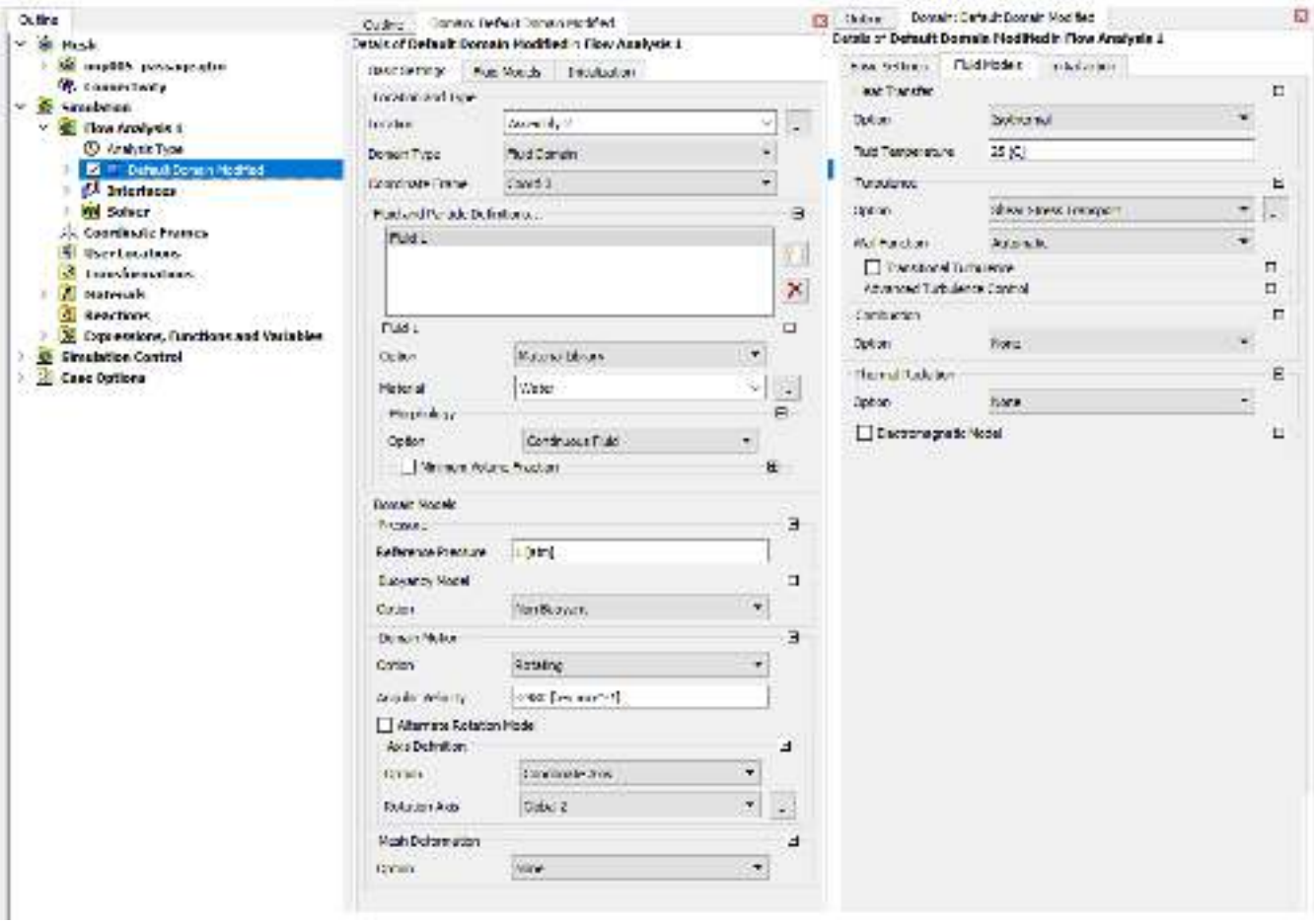


Figure 6 : Setting Model Fluid

- Create the blade setting as shown in the figure (see Figure 7), and with the same way set hub and shroud.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

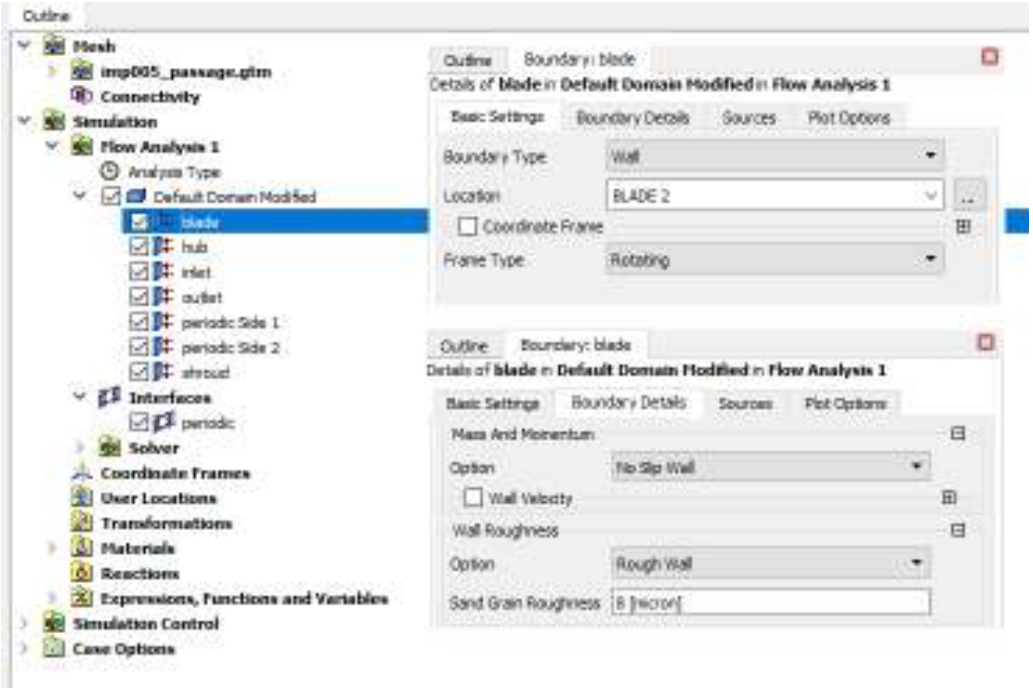


Figure 7 : Setting blade

- Create the inlet setting as shown in the figure (see Figure 8).

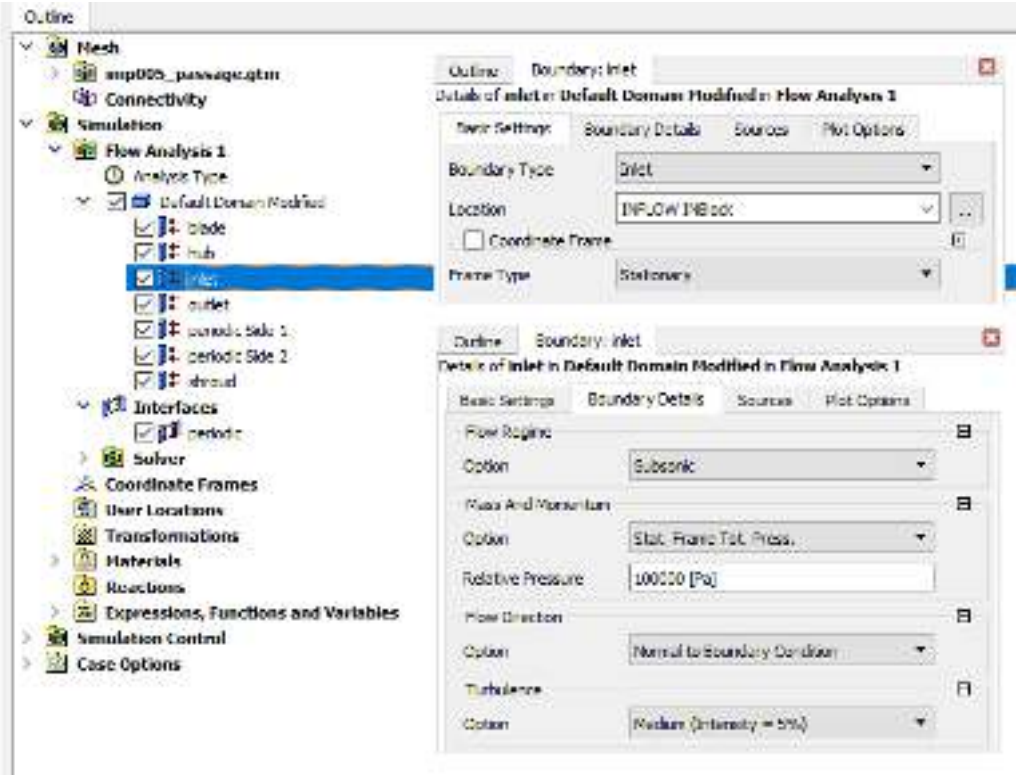


Figure 8 : Setting Inlet

- Create the outlet setting as shown in the figure (see Figure 9).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

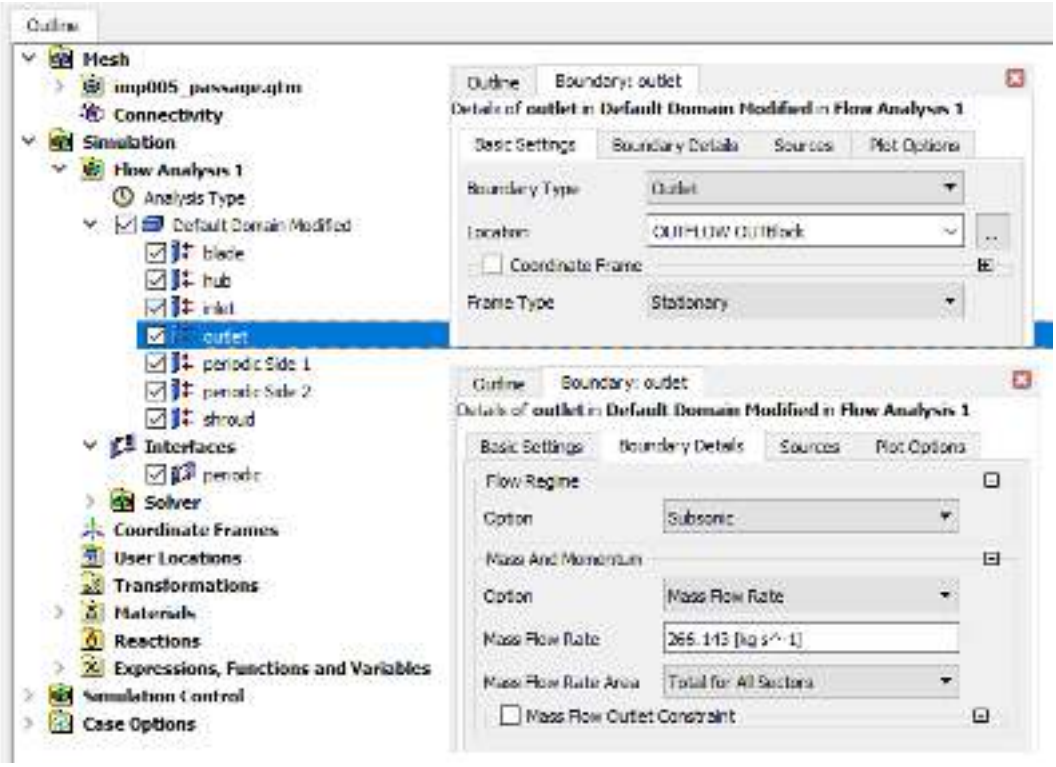


Figure 9 : Setting Outlet

- Create the periodic interface setting as shown in the figure (see Figure 10).

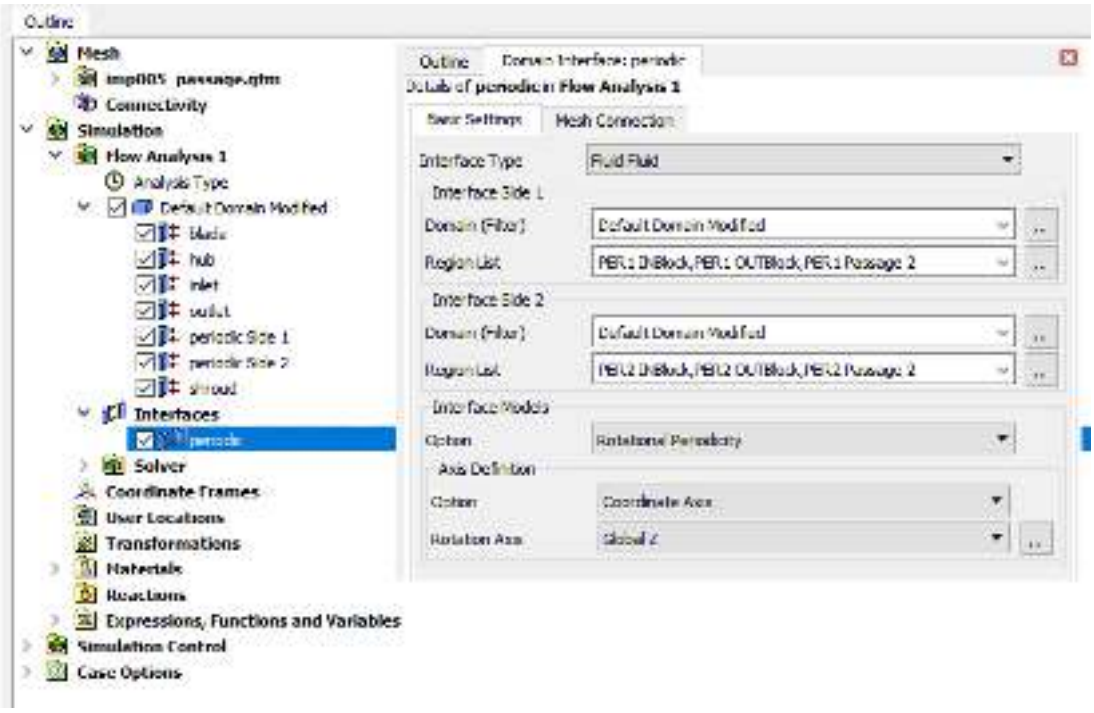


Figure 10 : Setting Periodic Interface

COPY COMPLIANT to the original document resident on the computer system at the time of printing

At the end of these first steps the domain of the impeller blades will assume the following configuration as shown in Figure 11:

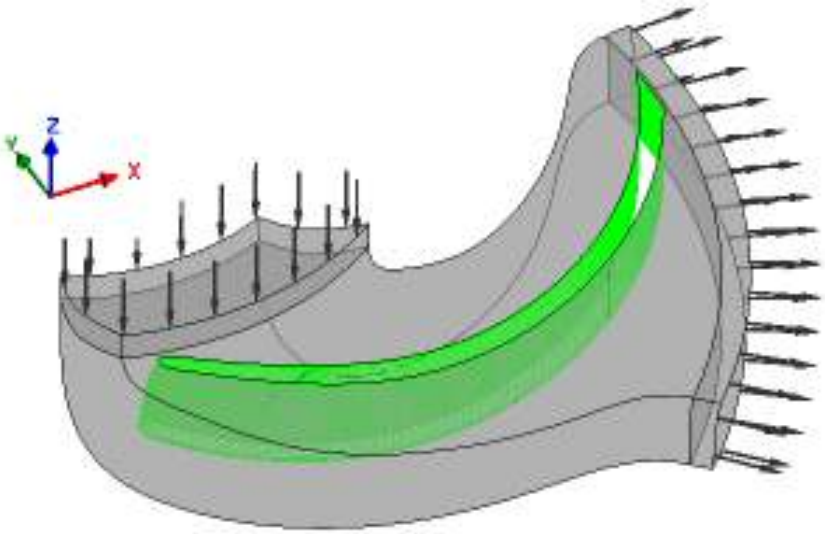


Figure 11 : Boundary Condition applied on vane domain

- Create the setting of “Solver Control” as shown in the figure (see Figure 12)

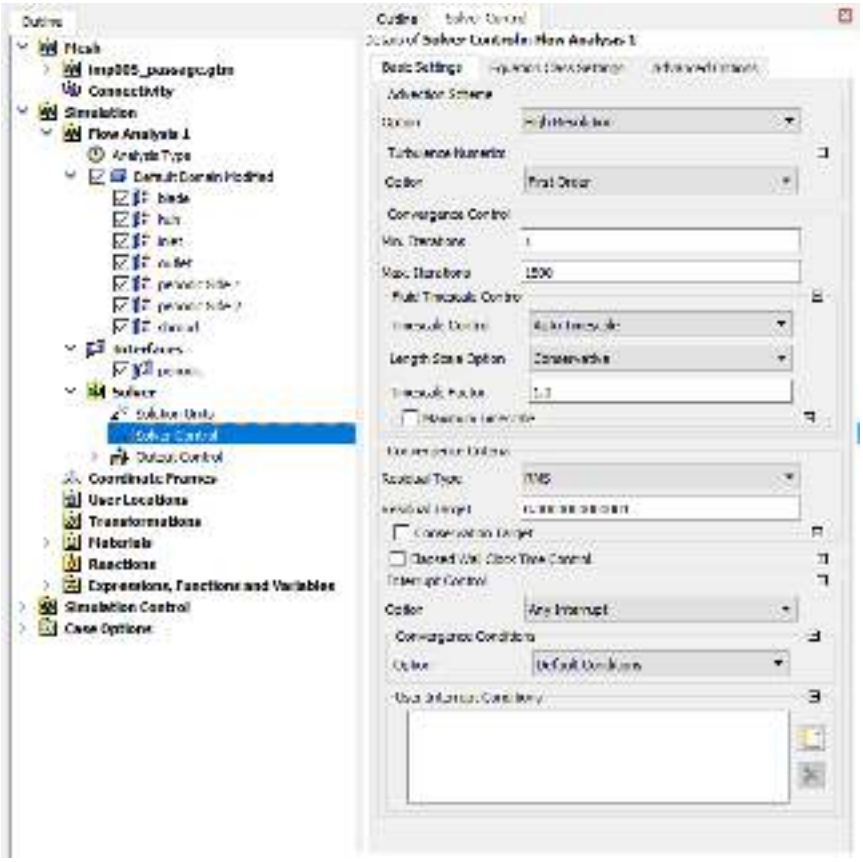


Figure 12 : Solver Control

COPY COMPLIANT to the original document resident on the computer system at the time of printing

- Create the Expressions as shown in the figure (see Figure 13)

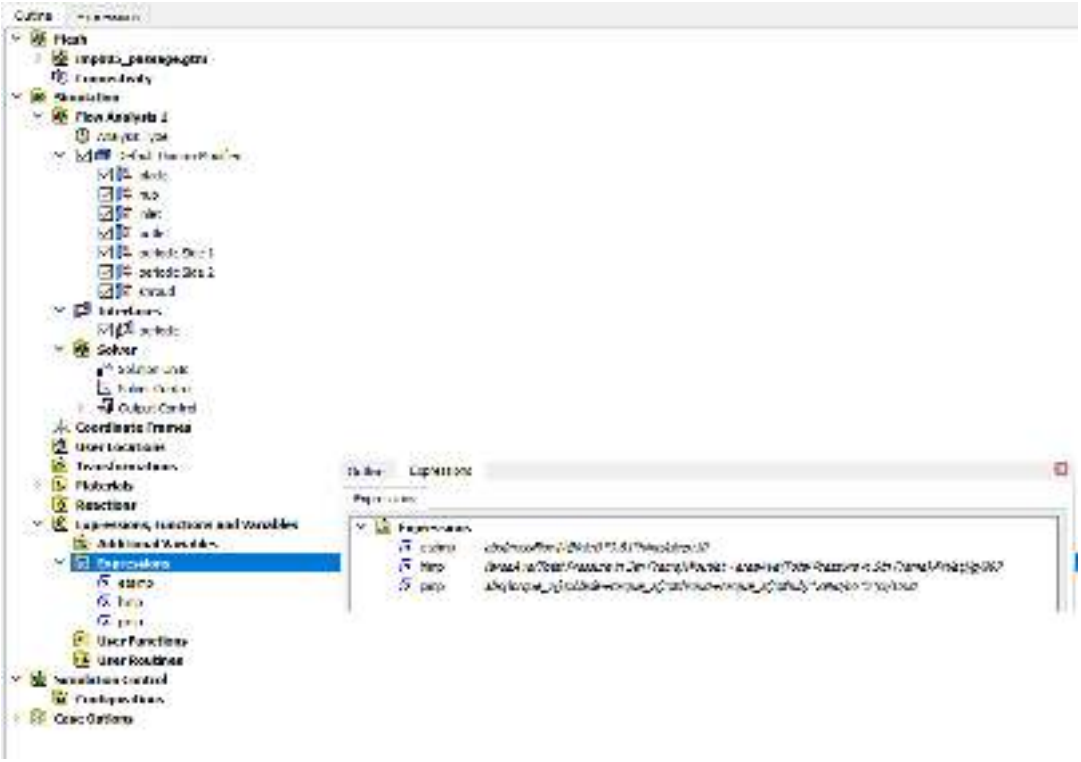


Figure 13 : Setting Expressions

- Create the Monitor for the expressions as shown in the figure (see Figure 14)

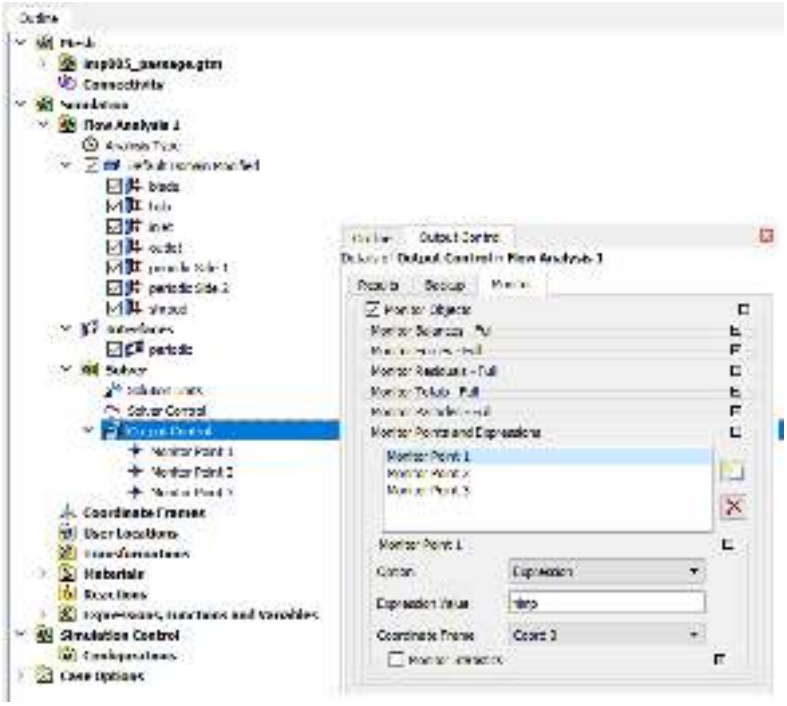


Figure 14 : Setting Monitor for Expressions

COPY COMPLIANT to the original document resident on the computer system at the time of printing

At the end of these steps save the solver input file with .def format as shown in Figure 15:

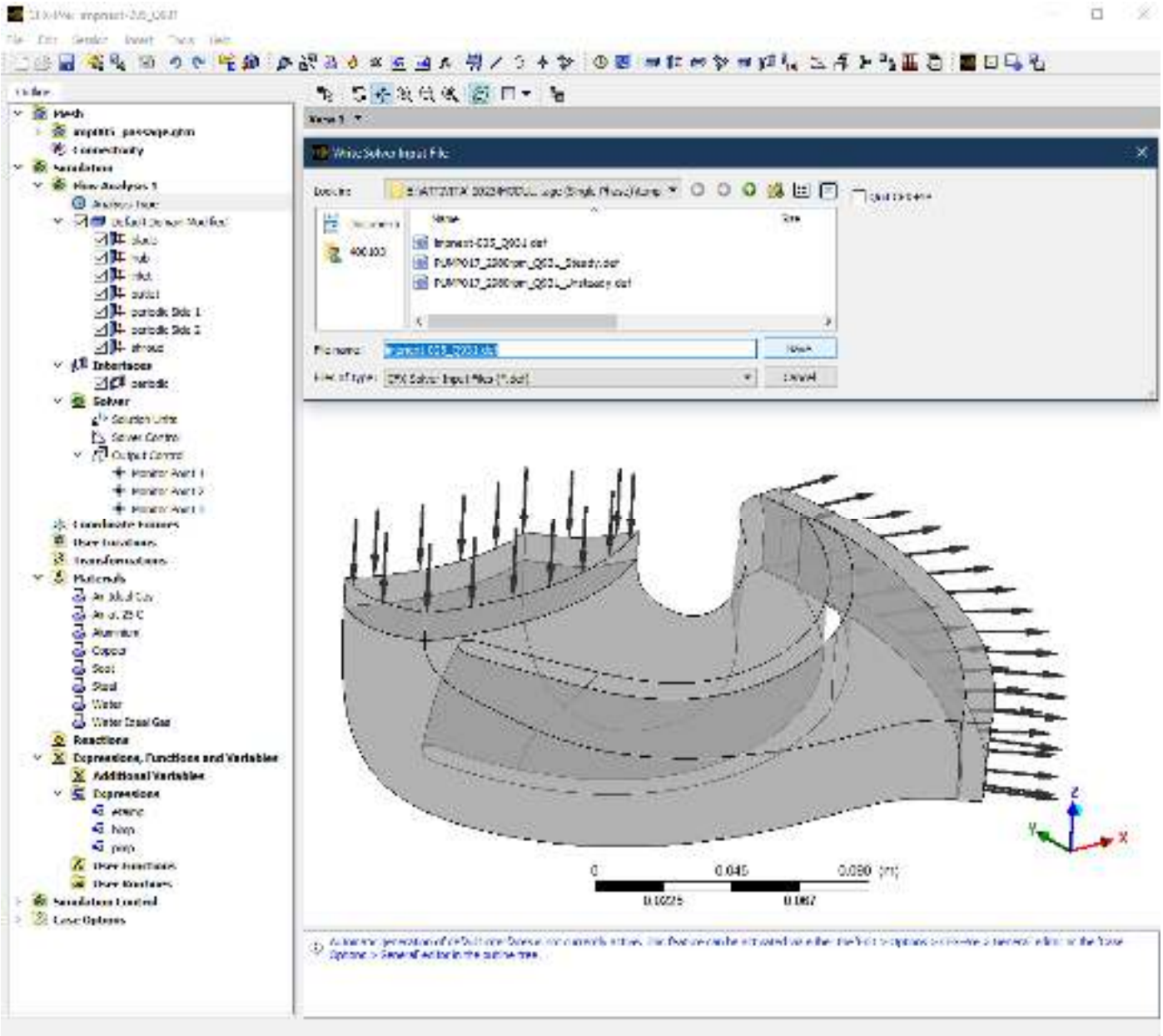


Figure 15 : Write Solver Input File

COPY COMPLIANT to the original document resident on the computer system at the time of printing